MATH0039 Differential and Integral Calculus

Year:	2026–2027
Code:	MATH0039
Level:	4 (UG)
Normal student group(s):	UG: Y1 Mathematics programmes
Value:	15 credits (7.5 ECTS credits)
Term:	1
Assessment:	85% examination, 15% coursework
Normal Pre-requisites:	A-level Maths or strong GCSE
Lecturer:	Dr T Gunaratnam, Dr Y Ma

Course Description and Objectives

This course provides a fairly rapid introduction to calculus. Calculus underlies almost all areas of mathematics and a great deal of science and engineering. The aim of the course will be to provide a solid grounding in this fundamental branch of mathematics for students who have a limited mathematical background.

Most of the course material occurs in A-level mathematics: the treatment here will be slightly more advanced. The course is suitable for students with good GCSE mathematics (or a weak A-level pass).

Recommended Texts

- Croft, Davison and Hargreaves, *Introduction to Engineering Mathematics* (Addison–Wesley)

Detailed Syllabus

1. Functions and graphs Polynomials, exponentials and logarithms, trigonometric functions. 2. Differentiation Differentiation from first principles, applications of differentiation, implicit differentiation, Taylor series. 3. Integration Fundamental theorem of calculus, indefinite and definite integrals, calculating area, integration by parts and substitution. 4. Differential equations First order differential equations, separation of variables and integrating factors, second order differential equations, complementary functions and particular integrals, numerical approximations, physical applications.

August 2026